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LFTSAD: Lightweight and Fast Time-Series Anomaly Detection

Using point-level and sequence-level reconstruction discrepancy

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Why this paper exists

The problem

Factories (Industry 4.0) stream sensor data. Anomalies must be caught in real time, on small edge devices such as a Raspberry Pi 4b (2 GB RAM). Deep models (Transformers, RNNs, CNNs) are too heavy and slow.

What exists

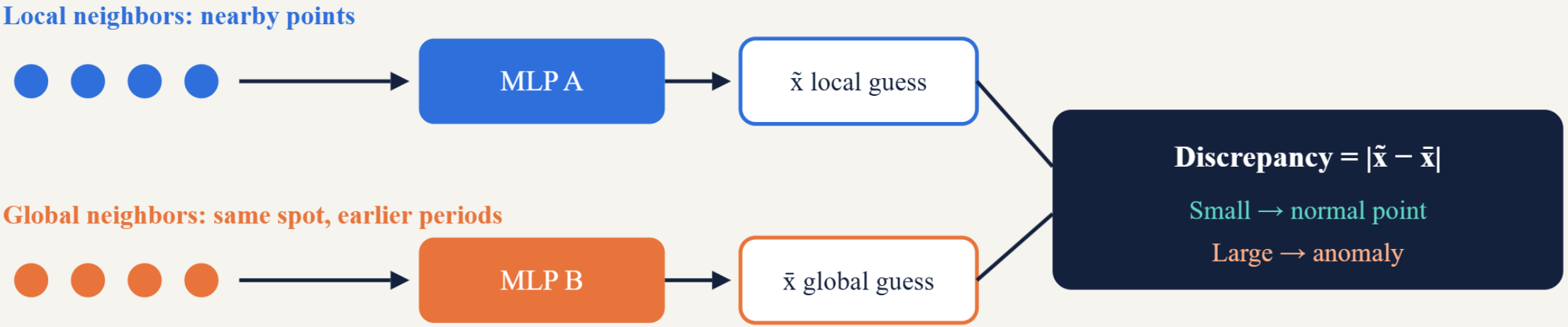
Anomaly Transformer and DCdetector are accurate, but compute costly "association" scores. Pruning or parallelism still keeps deep models. USAD and PatchAD use MLPs but are still deep.

Research gaps

- No shallow all-MLP detector built for edge devices
- Association scoring is too expensive
- Shallow nets miss subtle anomalies

Core idea: can the neighbors predict me?

Both networks try to rebuild the same timestamp. A normal point agrees with nearby and with distant, periodic points, so both guesses match. An anomaly only fits its own odd neighbors, so the guesses disagree.



How it works: four two-layer MLPs

Sensor data

Windows of the time series. Each variable is processed in parallel.

→

Point-level branch (MLP 1 and 2)

MLP 1 rebuilds a point from its local neighbors. MLP 2 rebuilds it from global neighbors. Score P is their difference.

Sequence-level branch (MLP 3 and 4)

Same trick for short sub-sequences instead of single points. Score S is their difference.

→

Final score

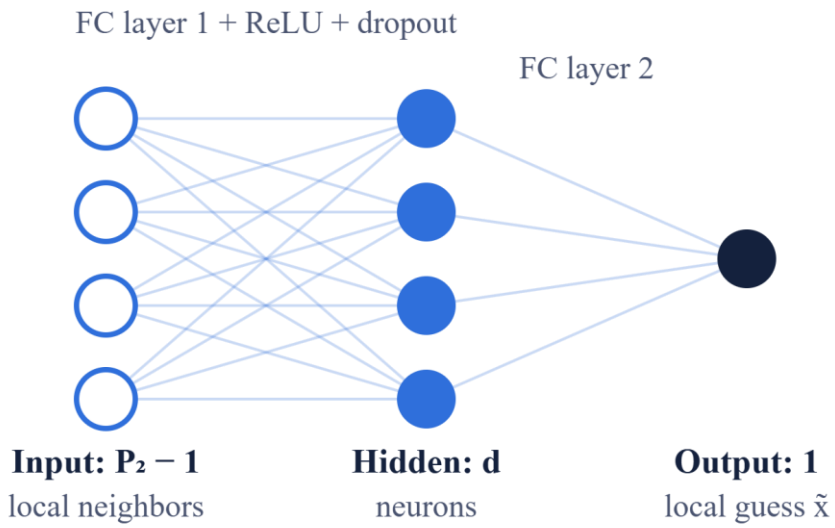
α·Score P + (1−α)·Score S

Score ≥ threshold λ means anomaly

Training: the loss (MSE) pulls the two guesses together on normal data. At test time, disagreement means anomaly. Only about 12.6K parameters.

Architecture of the point-level MLPs

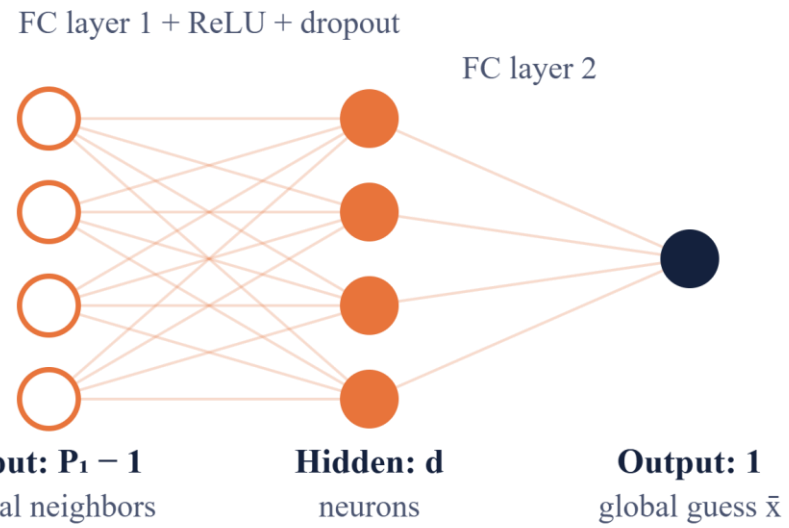
MLP 1: local association



Rebuilds one timestamp from the other points in its row.

Parameters: $(P_2-1)\times d + d\times 1$

MLP 2: global association

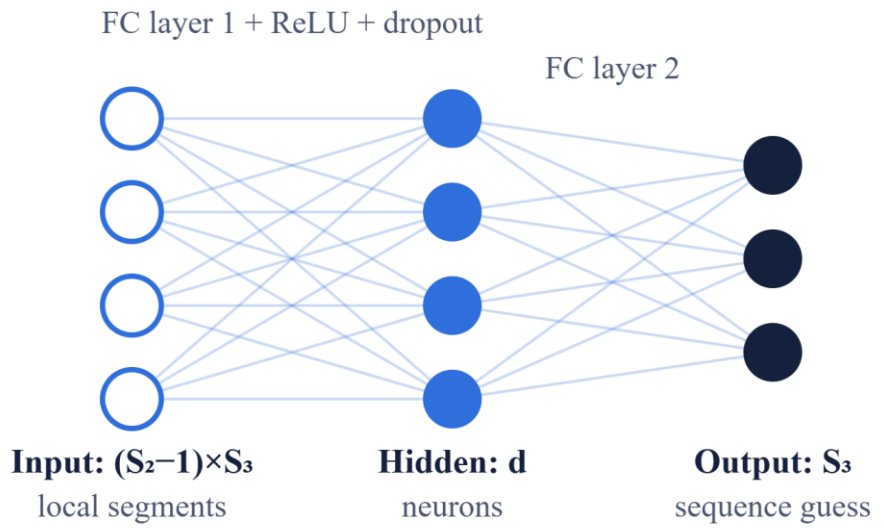


Rebuilds the same timestamp from points one period apart (its column).

Parameters: $(P_1-1)\times d + d\times 1$

Architecture of the sequence-level MLPs

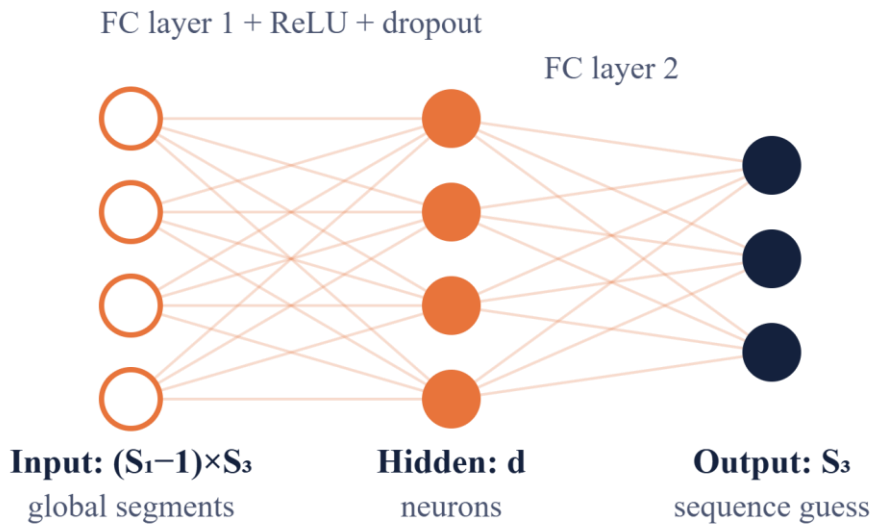
MLP 3: local association



Rebuilds a short substitute sequence of S_3 points from the local segments in its row.

Parameters: $(S_2-1)\times S_3\times d + d\times S_3$

MLP 4: global association

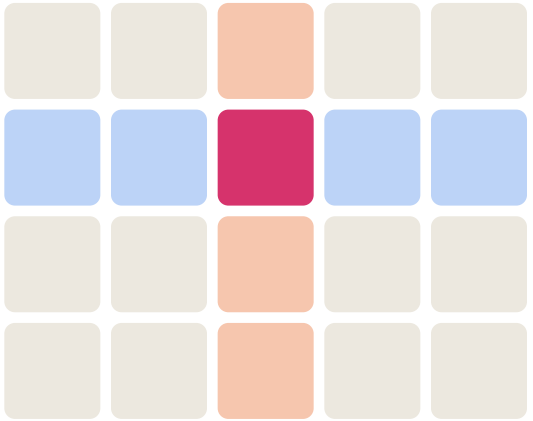


Rebuilds the same substitute sequence from segments one period apart (its column).

Parameters: $(S_1-1)\times S_3\times d + d\times S_3$

Why two levels? Explicit vs implicit anomalies

Neighbor sampling



A window is cut into a grid. Same row (blue) = local neighbors. Same column (orange) = global neighbors. Red = the point being tested.

Explicit outlier

Every anomalous point clearly differs from its neighbors. Point-level learning catches these well.

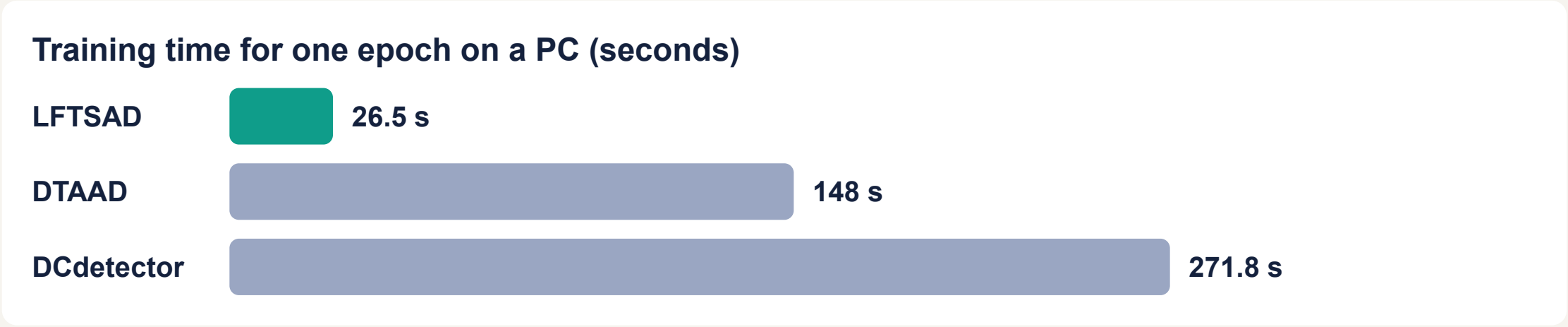
Implicit outlier

Some anomalous points look normal alone. Looking at a whole sub-sequence around the point exposes them. Sequence-level learning catches these.

Together

Joint learning beat either branch alone on every ablation dataset and metric reported.

Results: smaller, faster, comparably accurate



12.6K
parameters vs 4,741K for Anomaly Transformer

185 MB
RAM on Raspberry Pi 4b vs 402 MB for DCdetector

3–10×
faster than deep SOTA models (reported)

Top 2
on over half of 8 metrics across 14 datasets

Takeaways and limitations

Takeaways

- Compare two reconstructions instead of computing costly associations
- Four shallow MLPs are enough for edge deployment
- Point plus sequence level covers explicit and implicit anomalies

Limitations (from the paper)

- Sensitive to α on some datasets
- Does not model correlations between variables
- No online or federated updates yet

Future work

- Time-frequency learning, pruning, quantization
- Spatiotemporal joint learning
- Real-time online, federated system

Code: github.com/infogroup502/LFTSAD